

Getting Started With Raspberry Pi

The \$35 Raspberry Pi is a credit-card-size computer developed with the intention of promoting computer science. It is a pleasant exodus from the monotonous computing world. Here is how you can get your Raspberry Pi up and running

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Most computer science students get to learn high-level programming languages and application development but they don't really understand computers. The fun of controlling the real world with a real computer is never exposed to them.

A normal computer lacks the capability to communicate with the real world through real interfaces.

With a normal computer that has a monitor, keyboard, mouse, printer and modem connected, hardly anything can be done, except the intended use. Of course, the printer port on a PC can be used to interface with the real world but that will be very clumsy and cumbersome—not to talk of the non-availability of an easy programming environment for that.

The Raspberry Pi is a pleasant exodus from that monotonous computing world. You can hold this palm-size computer in your hand and at the same time interface with the real world with its 26 input/output (I/O) pins. With a Raspberry Pi in hand, robotics is no more a fancy imaginary world of big and serious people but a real and authentic one.

You can get this computer for a small amount of money. In the UK, the Raspberry Pi Model-A costs \$25, while the price of Model-B is \$35, plus local taxes and shipping/handling fees. In India, at many hobby outlets, it is available for only Rs 2600. In fact, the latest B Model with 512MB RAM is no less than a Celeron computer of yesteryear, yet with a whole lot of imagination and possibilities to tap.

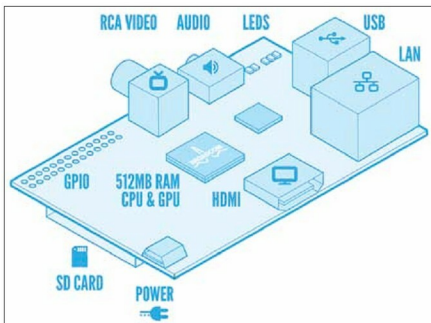


Fig. 1: Peripherals' connection (Courtesy: www.raspberrypi.org)



Fig. 2: Audio, HDMI, video cables

Model-A and Model-B

The first model of Raspberry Pi that appeared was Model-A, which had only one USB port and no Ethernet socket. Later, Model-B came into

market with two USB ports, Ethernet socket and 256MB RAM. The latest one has as much as 512MB RAM. It is built around a BCM2835 Broadcom processor. It consumes 5V electricity at 1A current. As power consumption